TITLE OF THE PROJECT

Project report submitted
in partial fulfilment of the requirements for the degree of

Bachelor of Technology in Mechatronics and Automation Engineering

by

Student Name (Reg. No)



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SCHOOL OF MECHANICAL ENGINEERING
VELLORE INSTITUTE OF TECHNOLOGY
CHENNAI

October, 2026

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DECLARATION

I/We, <1st Student Name (Reg. No), 2nd Student Name (Reg. No) and 3rd Student Name (Reg. No), B Tech Mechatronics and Automation Engineering, VIT Chennai, hereby declare that the work being presented in the dissertation entitled **"<PROJECT TITLE in Times New Roman, Font 12, Bold, CAPS >"** is an authentic record of the work that has been carried out at VIT, Chennai under the guidance of <Guide's name>, School of Mechanical Engineering, VIT Chennai.

I/We hereby declare that the entire work embodied in this dissertation has been carried out by me/us and no part of it has been submitted for any degree or diploma of any institution previously.

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CERTIFICATE

This is to certify that the project report entitled "**<THESIS TITLE in Times New Roman, Font 12, Bold, CAPS >"** submitted by <1st Student Name (Reg. No), 2nd Student Name (Reg. No) and 3rd Student Name (Reg. No), for the award of the degree of **Bachelor of Technology** is a record of bonafide work carried out by him/her/them under my supervision, as per the VIT code of academic and research ethics. The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university. The report fulfils the requirements and regulations of VIT and in my opinion meet the necessary standards for submission.

Signature of the Guide:

Name:

Date:

Signature of the Examiner-1

Name:

Date:

Signature of the Examiner-2

Name:

Date:

Approved by

Head of the Department

ABSTRACT

The abstract is a very brief summary of the project work. It should be about **200 to 250 words in length**. Abstracts contains a brief description of the problem and research objectives, method used, and the results of research. (typed one and a half line spacing, Font Style: Times New Roman and Font Size: 12, Paragraph alignment: justified)

Keywords: 3 to 5 key words which indicating salient features of the work.

(Example: Composite, Vibration, Tribology etc..)

ACKNOWLEDGEMENT

It is my pleasure to express with deep sense of gratitude to <Guide name>, <Designation>, School of Mechanical Engineering, Vellore Institute of Technology, Chennai, for his/her constant guidance, continual encouragement, understanding; more than all, he/she taught me patience in my endeavor. My association with him / her is not confined to academics only, but it is a great opportunity on my part of work with an intellectual and expert in the field of <area>.

It is with gratitude that I would like to extend my thanks to the visionary leader Dr. G. Viswanathan our Honorable Chancellor, Mr. Sankar Viswanathan, Dr. Sekar Viswanathan, Dr. G V Selvam Vice Presidents, Dr. Sandhya Pentareddy, Executive Director, Ms. Kadhambari S. Viswanathan, Assistant Vice-President, Dr. V. S. Kanchana Bhaaskaran Vice-Chancellor, Dr. T. Thyagarajan Pro-Vice Chancellor, VIT Chennai and Dr. P. K. Manoharan, Additional Registrar for providing an exceptional working environment and inspiring all of us during the tenure of the course.

In a jubilant state, I express my whole-hearted thanks to Dr. Karunamurthy K., Dean of School of Mechanical Engineering, Vellore Institute of Technology, Chennai, for spending their valuable time and efforts in sharing their knowledge and helping us in every aspect. Special mention also goes to Dr. Saravanakumar D, Head of the Department, B.Tech. Mechatronics and Automation Engineering and the Project Coordinators for their valuable support and encouragement to take up and complete the thesis.

My sincere thanks to all the faculties and staffs at Vellore Institute of Technology, Chennai who helped me acquire the requisite knowledge. I would like to thank my parents for their support. It is indeed a pleasure to thank my friends who encouraged me to take up and complete this task.

Place: Chennai

Date: Write student 1 Name (Register Number)

Write student 2 Name (Register Number)

Write student 3 Name (Register Number)

TABLE OF CONTENTS

Declaration pg. no. (roman)

Certificate pg. no. (roman)

Abstract pg. no. (roman)

Acknowledgement pg. no. (roman)

List of Figures pg. no. (roman)

List of Tables pg. no. (roman)

List of Nomenclature pg. no. (roman)

List of Acronyms and Keywords pg. no. (roman)

CHAPTER 1 pg. no. (1- page no of chapter end)

INTRODUCTION

1.1 INTRODUCTION 1

1.2 OVERVIEW OF MANET NETWORK 2

1.3 CHALLENGES PRESENT IN MANET 6

1.4 PROJECT STATEMENT 7

1.5 OBJECTIVES 7

1.6 SCOPE OF THE PROJECT 8

List all the title of chapter 1 in sequence

CHAPTER 2 pg. no.

REVIEW OF LITERATURE

List all the title of chapter 2 in sequence

CHAPTER 3 pg. no.

PROBLEM DEFINITION AND

OBJECTIVE

List all the title of chapter 3 in sequence

CHAPTER 4 pg. no.

METHODOLOGY

List all the title of chapter 4 in sequence

CHAPTER 5

RESULTS AND DISCUSSION

List all the title of chapter 5 in sequence

CHAPTER 6

CONCLUSION

List all the title of chapter 6 in sequence

REFERENCES

pg. no.

APPENDIX pg. no.

LIST OF PUBLICATION pg. no.

Exclude the APPENDIX and RESEARCH PUBLICATION sections from the index if they are blank.

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LIST OF FIGURES

Figure No.	Title	Page No.
2.1	Commonly used load-sensing drive schematic representation	9
2.2	Schematic representation of commonly used leakage compensation hydromotor drive	12
3.1	The schematic diagram of the test rig	37
3.2	Photographic view of the test rig	38
4.1	The schematic representation of a wind turbine with a hydraulic power transmission system	48
4.2	The experimental test rig used in the VIF study	49

Number each figure sequentially within its chapter.

LIST OF TABLES

Table No.	Title	Page No.
4.1	Design parameters of the flywheel	53
5.1	Comparison of the system control parameter for the constant speed drive of 1000rpm	87
6.1	Estimated PID parameters of the drives	99
6.2	Comparison of the system control parameter for the constant speed drive of 1000rpm	105
8.1	Spearman's correlation coefficient of the faulty parameter with the observer residual states	130
8.2	Pseudo-code of the RUL prediction	134
9.1	Comparison of performance parameters of five different hydromotor systems	140
A1.1	Generic Bond Graph elements with possible causal form	161
A1.2	Variables of power in different energy forms	162

Number each table sequentially within its chapter.

LIST OF NOMENCLATURE

I_{VIF}	Moment of inertia of the variable inertia flywheel
I_{FIF}	Moment of inertia of the fixed inertia flywheel
I_{dz}	Moment of inertia of the circular solid disk
I_{rz}	Moment of inertia of the removed slot material
I_{sz}	Moment of inertia of the slider
R	Outer radius of the flywheel
r	Inner radius of the flywheel
D_p	Displacement volume of the vane pump
D_m	Displacement volume of the bent axis hydromotor

The above is the sample.

LIST OF ACRONYMS AND KEYWORDS

HST	Hydrostatic transmission
PDCV	Proportional direction control valve
PRV	Pressure relief valve
PPRV	Proportional pressure relief valve
FCV	Flow control valve
SPC	Supply pressure compensator
PC	Pressure compensator of load-sensing Pump
FC	Flow compensator of load-sensing Pump
ANN	Artificial neural network
ANFIS	Adaptive neuro-fuzzy inference system
F2-PID	Fuzzy (type-2) proportional integral derivative
ARR	Analytical redundancy relation
UIO	Unknown input observer
NUIO	Nonlinear unknown input observer
FDI	Fault detection and isolation
RUL	Remaining useful life

KEYWORDS

Hydro-static Transmission System, Load-sensing hydromotor drive, Pump- controlled hydromotor drive, Valve-controlled hydromotor drive, Pump speed-controlled hydromotor drive, Active Controller, Passive Controller, Supply pressure compensator, PID Controller, Fuzzy type-2 PID Controller, Variable inertia flywheel, Fixed inertia flywheel, Fault diagnosis and prognosis, Unknown input observer, Remaining useful life prediction, degradation pattern, dSPACE Controller.

CHAPTER 1

INTRODUCTION

A Recipe for Writing an Introduction

An Introduction should contain the following three parts:

1.1 Background.

In this part you have to make clear what the context is. Ideally, you should give an idea of the state-of-the art of the field the report is about. But keep it short: in my opinion this part should be less than a page long. **About 200 words.**

1.2 The Problem.

If there was no problem, there would be no reason for writing a report, and definitely no reason for reading it. So, please tell the reviewer why should proceed reading. A simple sentence like "So far no-one has investigated the work..." or "The above-mentioned solutions don't apply to this case ...", can sometimes be enough to clarify the point you want to get at. Experience shows that for this part a few lines are often sufficient. **About 200 words.**

1.3 The Proposed Solution.

Now - and only now! - you may outline the contribution of the report. Here you have to make sure you point out what are the novel aspects of your work. There are probably millions of articles out there on that very subject: you can't expect the reviewer to know them all; so, make his life easier and clearly highlight what is the difference between your method and the others. You can take your time here, but I suggest to avoid getting into too much detail. **About 200 words.**

Include the necessary headings and subheadings, and ensure the numbering used is consistent in the Table of Contents (or index page)

CHAPTER 2

REVIEW OF LITERATURE

A literature review is a critical appraisal of the previous work published in the literature pertaining to the topic of the investigation. Literature reviews are secondary sources and do not report new or original experimental work. Critically read each source/journal paper, look for the arguments presented rather than for facts.

1. Narrow your topic and select papers accordingly

Consider your specific area of study. Think about what interests you and what interests' other researchers in your field. Limit your scope to a smaller topic area (ie. focusing on France's role in WWII instead of focusing on WWII in general).

2. Search for literature

Define your source selection criteria (ie. articles published between a specific date range, focusing on a specific geographic region, or using a specific methodology).

3. Read the selected articles thoroughly, Evaluate and synthesize the studies' findings and conclusions.

4. Minimum 20 recent journal papers need to cited and literature review should contain minimum of **600 to 1000 words in length.**

5. Provide a strong justification of the project's relevance to real-world industrial needs or societal challenges (Societal and Industry Significance)

6. Provide a brief comparison of alternative technologies considered and a preliminary cost-benefit analysis. Justify why the chosen approach is technically and economically feasible. **(Preliminary Cost and Feasibility Analysis)**

Include the necessary headings and subheadings, and ensure the numbering used is consistent in the Table of Contents (or index page)

CHAPTER 3

PROBLEM DEFINITION AND OBJECTIVES

3.1 Problem Definition

Must reflect the problem(s) identified from the review of literature. A problem statement is a concise description of the problem or issues a project seeks to address. The problem statement identifies the current state, the desired future state and any gaps between the two.
It can be represented in a paragraph.

3.2 Objective

The objectives that are achieved through this project can be listed in bullet form (maximum 4 bullet points). Concise and precise objective must be included. Wherever necessary paraphrase the objective and avoid redundancy.

1. Objective 1
2. Objective 2
3. ----------
4. ----------

3.3 Sustainability Goal

Explicitly state how these objectives align with specific Sustainability Goals

Include the necessary headings and subheadings, and ensure the numbering used is consistent in the Table of Contents (or index page)

CHAPTER 4

METHODOLOGY

In your dissertation or project report, you will have to discuss the methods you used to undertake your research. The methodology or methods section explains what you did and how you did it, allowing readers to evaluate the reliability and validity of your research.

Provide complete information about

- Materials used
- Equipment along with its technical specification
- Methods adopted for the study (experimental /numerical/coding, etc.)
- Insert mathematical equations using “Math equation” editor available in Ms-word

The methodology section should generally be written in the past tense **(about 500 words).**

Use flowchart to explain methodology of the project (for example refer methodology given below).

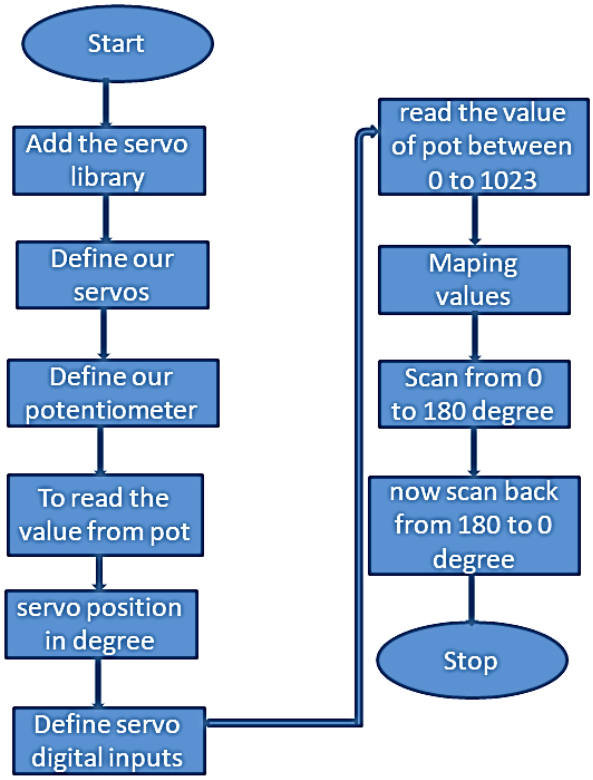


Figure 4.1 : Flow chart of methodology

Sample of the table:

S.No.	Case	Maximum Overshot (%)		Rise time		settling time	
		VIF	FIF	VIF	FIF	VIF	FIF
1	Case 1	6.57	15.71	5.51	5.51	6.31	6.57
2	Case 2	6.12	15.96	5.50	5.51	5.87	6.19
3	Case 3	5.37	16.54	5.48	5.50	5.77	6.17

Table 4.1 : Performance parameter of hydromotor drive with VIF and FIF

Include the necessary headings and subheadings, and ensure the numbering used is consistent in the Table of Contents (or index page)

CHAPTER 5

RESULTS AND DISCUSSION

The results section is a section containing a description about the main findings of a research, whereas the discussion section interprets the results for readers and provides the significance of the findings. Combining the results and discussion section allows for more coherence, because it allows you to discuss results of a particular test or method immediately after presenting them.

1. Complete description of work done
2. Use figures, flow charts, tables, videos, fabricated equipment/assembled components etc.
3. Provide proper discussion referring to literature
4. **About 500 to 1000 words**.

Innovation and Technical Depth: Explicitly highlight the "Innovation and Originality" of your solution. Discuss the technical complexity and how you optimized the system (e.g., code efficiency, mechanical weight reduction, or sensor accuracy).

Cost-Benefit and Sustainability Impact: Provide a final cost-benefit evaluation, comparing your prototype's cost against existing market solutions.

Environmental and Societal Impact: Discuss the project's environmental relevance and its scalability for future industrial or societal use. (Societal, Industrial & Future Scope Impact)

Include the necessary headings and subheadings, and ensure the numbering used is consistent in the Table of Contents (or index page)

CHAPTER 6

CONCLUSION

- Summarize the core achievements and state clearly how each of the initial objectives was successfully met. Give concise and clear conclusion in 3 – 4 bullet points
- Wherever necessary paraphrase the conclusion and avoid redundancy.
- Discuss how this project can be scaled for mass production or adapted for different industrial sectors (Societal and Industrial Scalability)
- Describe at least two major technical hurdles or "errors" encountered during the project (e.g., sensor calibration issues, mechanical failure, or code bugs). Explain the steps taken to troubleshoot and solve these issues.
- Suggest specific features or technical upgrades that could be implemented in future work of the project

About 300 words

.

REFERENCES

(Use continuous number format throughout the reference section. Follow the reference templates for ***Journal publication, Conference proceedings, web reference, book*** as given below. format for entire references).

Journal Publications:

1. Duzdevich, D., Redding, S. and Greene, E. C. (2014), 'Dna dynamics and Single-Molecule Biology', *Chemical Reviews* **114**(6), 3072–3086.
2. Maedche, A. and Staab, S. (2001), 'Ontology learning for the semantic web', *IEEE Intelligent Systems* **16**(2), 72–79.
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Conference Proceedings:

4. Doan, A., Madhavan, J., Domingos, P. and Halevy, A. (2002), Learning to map between ontologies on the semantic web, *in* 'Proceedings of the 11th international conference on World Wide Web', ACM, pp. 662–673.

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5. Gilbarg, D. and Trudinger, N. S. (2015), '*Elliptic partial differential equations of second order*', Springer Publications.
6. Kothari, C. R. (2004), '*Research methodology: Methods and techniques*', New Age International Publications.

Websites:

7. Google Maps. (2019, February 5). *The British Library, London, UK*. Google. Retrieved from https://www.google.com/search?q=scope+meaning&oq=scope+&aqs=chrome.6.69i57j0l2j0i433j0j0i433l3j0l2.4838j0j15&sourceid=chrome&ie=UTF-
8. Flower, R. (2019, June 1). How a simple formula for resolving problems and conflict can change your reality [Blog post]. Pick The Brain. Retrieved from http://www.pickthebrain.com/blog/how-a-simple-formula-for-resolving-problems-and-conflict-can-change-your-reality/

APPENDIX A

It is essential to consult with your project guide to confirm the inclusion of these supplementary materials. Propose organizing each item such as the introduction on simulation tools, working details of major components, sensor specifications, algorithm codes (Python/MATLAB) and Plagiarism Report —as a separate, clearly labeled appendix (e.g., Appendix A, Appendix B).

The appendix section is used for details that are important but don't fit into your main chapters. Based on your project's specific requirements, your guide may also suggest including other relevant appendices, such as raw data tables, detailed calculations, or component datasheets. This ensures your final report is comprehensive and professionally documented.

Exclude this section if it is blank.

LIST OF PUBLICATION

Published Journal

1. **Authors**. "A comparative analysis of the pump controlled, valve controlled and prime mover controlled hydromotor drive to attain constant speed for varying load." ISA transactions (2021). (SCI/Q1/5.9)
2. **Authors**. "Dynamic analysis of a hydraulic motor drive with variable inertia flywheel." *Proceedings of the Institution of Mechanical Engineers, Part I: Journal of Systems and Control Engineering* (2019): 0959651819875914. (SCI/Q4/1.6)
3. **Authors**. "Model-Based Performance Analysis of a Hybrid Summation Drive Used in Off-Highway Vehicles." *Arabian Journal for Science and Engineering* 44.2 (2019): 857-872. (SCI/Q2/2.8)

Communicated Journal

1. **Authors** " *Fault diagnosis and prognosis of hydraulic drive components based on unknown input observer and Takagi- Sugeno fuzzy method*." ISA transactions".
2. **Authors** " *A comparative analysis of the load-sensing hydromotor drives to maintain a constant flow under varying load conditions*." Journal of Dynamic Systems, Measurement, and Control: Transaction of the ASME".

Scopus Journals

1. **Authors** "Bond graph modelling and simulation of a variable inertia flywheel." *IOP Conference Series: Materials Science and Engineering*. Vol. 377. No. 1. IOP Publishing, 2018.

To document your research output, list the full citation for each article (published/communicated). For published papers, you must attach a clear image of the article's first page as proof. For articles that have been submitted but are not yet published, attach a screenshot of the official submission confirmation email or portal page as evidence.

Exclude this section if it is blank.